E-Waste Comparison and Rules of Different Countries

Comparison of China, India, Russia, Thailand and Singapore

1. E-Waste Generation Comparison (2022)

Country	E-waste generated	Per capita	Overall situation
CN **China**	≈12.1 million tonnes	≈8.5 kg/person	Highest total among these countries
IN **India**	≈4.1 million tonnes	≈2.9 kg/person	Very high total, but low per-person generation
RU **Russia**	≈1.6 million tonnes	≈11–12 kg/person	High per-person generation
TH **Thailand**	≈0.75 million tonnes	10.5 kg/person	High per-person generation
SG **Singapore**	≈0.12 million tonnes	≈20 kg/person	Very high per-person generation

China generated about 12 million tonnes of e-waste in 2022, while India generated about 4.1 million tonnes. Thailand generated about 753,000 tonnes (10.5 kg/person). Singapore generated a much smaller total because of its population, but its per-capita generation was around 20 kg/person, among the highest in Asia.

1.1 Multi-Year E-Waste Generation (Million Tonnes), 2019–2024

Country	2019	2020	2021	2022
CN **China**	10.23	10.85	≈11.40	12.07
IN **India**	≈3.23	3.35	≈3.80	4.14
RU **Russia**	≈1.63	≈1.66	≈1.69	≈1.60
TH **Thailand**	0.678	≈0.70	≈0.72	0.753
SG **Singapore**	0.114	0.117	0.119	0.121

Unit: Million tonnes (Mt). 2023–2024 figures not yet available (N/A) for these countries.

1.2 Multi-Year E-Waste Per Capita (kg/person), 2019–2022

Country	2019	2020	2021	2022	Trend
CN China	≈7.2	≈7.6	≈8.0	8.5	Increasing
IN India	≈2.4	2.4	≈2.7	2.9	Increasing
RU Russia	≈11.3	≈11.5	≈11.7	≈11–12	High / Stable
TH Thailand	≈9.5	≈9.8	≈10.1	10.5	Increasing
SG Singapore	≈19.6	≈19.8	≈20.1	≈20.0	Very High

2. What the Comparison Tells Us

China → Largest e-waste problem by total quantity. Its huge population and high electronics consumption result in enormous e-waste generation.

India → Second largest in this comparison by total quantity, but per-capita generation is much lower than the other four countries.

Russia → High per-capita generation, reflecting relatively high consumption of electrical and electronic equipment.

Thailand → Medium total quantity but high per-capita generation at 10.5 kg/person. Only about 29–32 thousand tonnes was formally collected in 2022.

Singapore → Smallest total quantity but the highest per-capita level among these five, around 20 kg/person. This is associated with a high-income, highly digitized economy and rapid replacement of electronics.

3. Ranking

By total e-waste: China > India > Russia > Thailand > Singapore

By e-waste per person: Singapore > Russia ≈ Thailand > China > India

The key point is that China and India have the biggest challenge in terms of total e-waste volume, while Singapore has the biggest challenge per person.

4. Rules and E-Waste Management Comparison

Country	Main regulatory approach	Important focus
CN China	National e-waste management framework with producer responsibilities and regulated recycling channels.	Formal collection, recycling capacity, pollution control and resource recovery.

Country	Main regulatory approach	Important focus
IN India	Extended Producer Responsibility (EPR) under the E-Waste (Management) Rules, with producers/importers/manufacturers responsible for meeting recycling obligations.	Increasing formal recycling, EPR targets, authorized recyclers and reduction of informal handling.
RU Russia	Producer/importer responsibility combined with national waste-management and extended producer responsibility mechanisms.	Collection, recycling obligations and development of formal waste-management infrastructure.
TH Thailand	E-waste is managed through environmental and hazardous-waste controls, with increasing emphasis on formal collection and recycling systems.	Reducing informal processing, improving collection and safe recycling.
SG Singapore	A regulated e-waste system based on producer responsibility, with the Resource Sustainability Act establishing an EPR framework for regulated products.	Convenient collection points, licensed recycling, traceability and high-quality resource recovery.

5. Major Similarities in the Five Countries

- All five countries recognize e-waste as an environmental and resource-management issue.
- All are moving toward more formal collection and recycling systems rather than uncontrolled disposal.
- Producer responsibility is an important policy direction, although the exact implementation differs by country.
- Safe handling of hazardous substances and recovery of valuable materials such as metals are common objectives.
- A major challenge is improving collection rates and reducing informal or environmentally unsafe recycling practices.

6. Overall Conclusion

The comparison shows that e-waste generation depends strongly on both population size and consumption patterns. China and India create the largest total quantities because of

their large populations, while Singapore, Russia and Thailand have much higher per-capita generation. Therefore, countries need both large-scale recycling infrastructure and strong per-person waste-reduction measures. Strengthening EPR, formal collection, authorized recycling and resource recovery can help reduce environmental impacts and improve the recovery of valuable materials.

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